

DAY 8 MACHINE TEST – Linux Troubleshooting & Real DevOps Commands

Junior DevOps Engineer Interview-Oriented Assignment

Objective

You have joined a company as a Junior DevOps Engineer. Multiple incidents are reported from production servers. Your task is to investigate issues, collect evidence, identify root causes, and document findings using Linux troubleshooting commands learned during training.

PART 1 – Linux Fundamentals & Server Health

Check

1. Explain whoami.

whoami displays the username of the currently logged-in user.

2. Explain hostname.

hostname displays the name of the computer/server on which you are currently working.

3. Why is uptime important during troubleshooting?

uptime shows:

- How long the system has been running
- Number of logged-in users
- System load average

4. Explain df -h and its DevOps usage.

df -h displays disk space usage of mounted filesystems in a human-readable format. DevOps usage: It is used to monitor disk space and identify full disks, which can cause application or service failures.

5. Explain free -h and its output.

free -h displays RAM and swap memory usage in human-readable format.

6. Explain top and its importance.

top displays real-time system and process information.

7. Explain systemctl --failed.

systemctl --failed displays systemd services that have failed. It is useful for quickly identifying failed services that may be causing application or server problems.

8. Explain journalctl -p err -n 20 and its usage.

This command displays the latest 20 error-level messages from the system journal.

- **journalctl → View system logs**

- -p err → Show error-level logs
- -n 20 → Show the latest 20 entries

Usage: It is useful for quickly checking recent system errors during troubleshooting.

PART 2 – Production Troubleshooting Scenarios

Scenario 1 – High CPU Usage

Users report the application is slow. Investigate CPU consumption, identify processes, collect evidence, explain corrective actions and workflow.

Workflow:

Check CPU → Identify process → Collect evidence → Find root cause → Take corrective action → Verify → Document

1) uptime

2)top

3) ps aux --sort=--%cpu | head

Scenario 2 – High Memory Usage

Investigate RAM pressure, identify memory-consuming processes and propose remediation.

1) free -h

2) ps aux --sort=--%mem | head

3) Top

Workflow:

Check memory → Identify process → Check logs → Find root cause → Remediate → Verify → Monitor

Scenario 3 – Disk Full

Investigate a deployment failure caused by 100% disk usage. Identify root cause, corrective actions and preventive measures.

1) df -h

Scenario 4 – Website Down

Investigate Nginx service issues, restart procedures, verification steps and explain the role of Nginx.

Nginx is a web server and reverse proxy used to serve websites and forward requests to backend applications.

1) Sudo systemctl status nginx

2) Sudo nginx -t

3) Sudo systemctl restart nginx

4) Sudo systemctl status nginx

5) Sudo journalctl -u nginx -n 50

Workflow:

Check service → Test configuration → Check logs → Restart if required → Verify → Test website

Scenario 5 – Application Crash

Use logs to identify root cause and document findings.

- 1) `sudo systemctl status <application-service>`
- 2) `sudo journalctl -u <application-service> -n 50`

I look for error messages related to configuration, memory, permissions, database connections, ports, or missing files.

Scenario 6 – Permission Denied

Investigate file permissions, ownership and executable permissions.

- 1) `ls -l <file>`
- 2) `ls -l script.sh`
- 3) `chmod +x script.sh`
- 4) `sudo chown user:group <file>`

Workflow:

Check permissions → Check ownership → Check execute permission → Fix required permission → Verify → Test

PART 3 – Practical Linux Tasks

Task 1: Capture output of whoami, hostname and uptime.

 sreelekha@LAPTOP-CKI0PC9R: ~

```
sreelekha@LAPTOP-CKI0PC9R:~$ whoami
sreelekha
sreelekha@LAPTOP-CKI0PC9R:~$ hostname
LAPTOP-CKI0PC9R
sreelekha@LAPTOP-CKI0PC9R:~$ uptime
11:15:02 up 55 min, 1 user, load average: 0.01, 0.04, 0.01
sreelekha@LAPTOP-CKI0PC9R:~$
```

Task 2: Collect output of free -h, top and df -h.

df -h

```
sreelekha@LAPTOP-CKI0PC9R:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G   0 1.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            1.9G  4.0K 1.9G   1% /mnt/wsl
drivers         205G  163G   43G  80% /usr/lib/wsl/drivers
/dev/sdd        1007G   2.4G  954G   1% /
none            1.9G  40K 1.9G   1% /mnt/wslg
none            1.9G   0 1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G   2.8M 1.9G   1% /init
none            1.9G  556K 1.9G   1% /run
none            1.9G   0 1.9G   0% /run/lock
none            1.9G   0 1.9G   0% /run/shm
none            1.9G   80K 1.9G   1% /mnt/wslg/versions.txt
none            1.9G   80K 1.9G   1% /mnt/wslg/doc
C:\             205G  163G   43G  80% /mnt/c
D:\             138G   3.2G  135G   3% /mnt/d
E:\             135G   2.2G  133G   2% /mnt/e
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.9G   0 1.9G   0% /tmp
none            1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service
tmpfs           374M  12K  374M   1% /run/user/1000
```

Top

```
top - 11:16:19 up 56 min, 1 user, load average: 0.00, 0.03, 0.00
Tasks: 37 total, 1 running, 36 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 3737.6 total, 3053.0 free, 479.9 used, 282.7 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used, 3257.7 avail Mem

  PID USER      PR  NI   VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24704  15924  11664 S   0.0   0.4   0:01.00 systemd
    2 root        20   0   3180   2288   2072 S   0.0   0.1   0:00.00 init-systemd(Ub
    8 root        20   0   3180   2040   1940 S   0.0   0.1   0:00.00 init
   49 root        19  -1  50404  17024  15708 S   0.0   0.4   0:00.20 systemd-journal
   81 systemd+    20   0   22416  14528  12032 S   0.0   0.4   0:00.07 systemd-resolve
   88 root        20   0   35280  12284   9200 S   0.0   0.3   0:00.27 systemd-udev
  110 root        20   0   2888   2076   1940 S   0.0   0.1   0:00.06 chronyd-starter
  111 root        20   0   4472   3236   2080 S   0.0   0.1   0:00.01 cron
  112 message+    20   0   8856   5516   4792 S   0.0   0.1   0:00.12 dbus-daemon
  145 root        20   0   44100  29916  14768 S   0.0   0.8   0:00.23 networkd-dispat
  154 root        20   0   13440   9508   8260 S   0.0   0.2   0:00.11 systemd-logind
  189 syslog      20   0  220548   6016   4704 S   0.0   0.2   0:00.08 rsyslogd
  191 root        20   0   14856   2516   1036 S   0.0   0.1   0:00.00 nginx
  192 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  193 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  194 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  196 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  197 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  198 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  199 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  200 www-data     20   0   16904   5288   3428 S   0.0   0.1   0:00.00 nginx
  218 root        20   0   5492   2848   2660 S   0.0   0.1   0:00.00agetty
  276 _chrony     20   0   23600  11168   7192 S   0.0   0.3   0:00.06 chronyd
  293 _chrony     20   0   12096   2500   1856 S   0.0   0.1   0:00.00 chronyd
  318 root        20   0  123464  32788  18180 S   0.0   0.9   0:00.18 unattended-upgr
  320 root        20   0  1940944  40976  26188 S   0.0   1.1   0:04.10 containerd
  389 root        20   0  2203000  79896  58192 S   0.0   2.1   0:00.80 dockerd
  643 root        20   0   8648   5588   4744 S   0.0   0.1   0:00.00 login
  685 sreelek+    20   0   22360  13684  11048 S   0.0   0.4   0:00.10 systemd
  687 sreelek+    20   0   22920   4120   2116 S   0.0   0.1   0:00.00 (sd-pam)
  715 sreelek+    20   0   6136   5564   3992 S   0.0   0.1   0:00.02 bash
 1134 polkitd     20   0  306448   8516   7696 S   0.0   0.2   0:00.15 polkitd
 1586 root        20   0  1857908  14056  11380 S   0.0   0.4   0:00.04 wsl-pro-service
```

Free -h

```
sreelekha@LAPTOP-CKI0PC9R:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          3.7Gi         505Mi        3.0Gi         3.5Mi         282Mi        3.2Gi
Swap:          1.0Gi           0B          1.0Gi

sreelekha@LAPTOP-CKI0PC9R:~$
```

Task 3: Identify top CPU-consuming processes.

```
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux --sort=-%cpu | head
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           320  0.1  1.0 1940944 40980 ?        Ssl  10:19   0:04 /usr/bin/containerd
polkitd       1134  0.0  0.2 306448  8516 ?        Ssl  11:10   0:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=noti
ce
root           1  0.0  0.4 24704 15924 ?        Ss   10:19   0:01 /sbin/init
root          389  0.0  2.0 2203000 79896 ?        Ssl  10:19   0:00 /usr/bin/dockerd -H fd:// --containerd=/run/container
d/containerd.sock
sreelekha+   1889  0.0  0.1  6268  5664 pts/0    Ss   11:13   0:00 -bash
root         1586  0.0  0.3 1857908 14056 ?        Ssl  11:10   0:00 /usr/libexec/wsl-pro-service
root          88  0.0  0.3  35280 12204 ?        Ss   10:19   0:00 /usr/lib/systemd/systemd-udevd
root         145  0.0  0.7 44100 29916 ?        Ss   10:19   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
tartup-triggers
root          49  0.0  0.4  50404 17036 ?        S<s  10:19   0:00 /usr/lib/systemd/systemd-journald
```

Task 4: Identify top memory-consuming processes.

```
sreelekha@LAPTOP-CKI0PC9R:~$ ps aux --sort=-%mem | head
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          389  0.0  2.0 2203000 79960 ?        Ssl  10:19   0:00 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
root          320  0.1  1.0 1940944 40980 ?        Ssl  10:19   0:04 /usr/bin/containerd
root          318  0.0  0.2 123464  32788 ?        Ssl  10:19   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root         145  0.0  0.7 44100 29916 ?        Ss   10:19   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root          49  0.0  0.4  50404 17036 ?        S<s  10:19   0:00 /usr/lib/systemd/systemd-journald
root           1  0.0  0.4 24704 15924 ?        Ss   10:19   0:01 /sbin/init
systemd+      81  0.0  0.3 22416 14528 ?        Ss   10:19   0:00 /usr/lib/systemd/systemd-resolved
root         1586  0.0  0.3 1857908 14056 ?        Ssl  11:10   0:00 /usr/libexec/wsl-pro-service
sreelekha+   685  0.0  0.3 22360 13684 ?        Ss   10:19   0:00 /usr/lib/systemd/systemd --user
sreelekha@LAPTOP-CKI0PC9R:~$
```

Task 5: List failed services.

```
sreelekha@LAPTOP-CKI0PC9R:~$ systemctl --failed
UNIT LOAD ACTIVE SUB DESCRIPTION
0 loaded units listed.
sreelekha@LAPTOP-CKI0PC9R:~$
```

Task 6: View latest error logs.

```
sreelekha@LAPTOP-CKI0PC9R:~$ journalctl -p err -n 20
Oct 07 09:34:06 LAPTOP-CKI0PC9R unknown: WSL (134) ERROR: CheckConnection: getaddrinfo() failed: -5
-- Boot b85849bf23ff4e7c8ace9cea65ae64bd --
Oct 07 15:49:02 LAPTOP-CKI0PC9R kernel: PCI: Fatal: No config space access function found
Oct 07 15:49:02 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
Oct 07 15:49:02 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
Oct 07 15:49:02 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
Oct 07 15:49:02 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
Oct 07 15:49:02 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
Oct 07 15:49:11 LAPTOP-CKI0PC9R chronyd[288]: Could not step system clock
Oct 07 15:49:12 LAPTOP-CKI0PC9R chronyd[288]: Could not step system clock
Oct 07 15:49:14 LAPTOP-CKI0PC9R chronyd[288]: Could not step system clock
Oct 07 15:49:46 LAPTOP-CKI0PC9R unknown: WSL (136) ERROR: CheckConnection: getaddrinfo() failed: -5
Oct 07 16:29:23 LAPTOP-CKI0PC9R systemd[1]: Failed unmounting init.mount - /init.
Oct 07 16:29:23 LAPTOP-CKI0PC9R systemd[1]: Failed unmounting tmp-.X11\x2dunix.mount - /tmp/.X11-unix.
-- Boot 365a94c4eba94486bfd528fdaa87a5b3 --
Oct 08 10:19:25 LAPTOP-CKI0PC9R kernel: PCI: Fatal: No config space access function found
Oct 08 10:19:25 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
Oct 08 10:19:25 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
Oct 08 10:19:25 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
Oct 08 10:19:25 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
Oct 08 10:19:25 LAPTOP-CKI0PC9R kernel: misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
Oct 08 10:20:12 LAPTOP-CKI0PC9R unknown: WSL (135) ERROR: CheckConnection: getaddrinfo() failed: -5
sreelekha@LAPTOP-CKI0PC9R:~$
```

Task 7: Find the largest directory in the home folder.

```
sreelekha@LAPTOP-CKI0PC9R:~$ du -sh ~/* | sort -hr | head
32K    /home/sreelekha/linux_test
n28K    /home/sreelekha/linux-test
16K     /home/sreelekha/devops
16K     /home/sreelekha/content
16K     /home/sreelekha/LinuxPractice
12K     /home/sreelekha/project-backup.tar
8.0K    /home/sreelekha/storage-report.txt
8.0K    /home/sreelekha/project
8.0K    /home/sreelekha/mount-report.txt
8.0K    /home/sreelekha/devOps
sreelekha@LAPTOP-CKI0PC9R:~$
```